

Problem-Solution Fit

Date	05 July 2026
Team ID	SWTID-2026-4324
Project Name	My project is opticrop: smart agricultural production optimization engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Small and medium-scale farmers - Large commercial farms - Agricultural cooperatives - Agri-business companies - Government agriculture departments	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited budget - Low digital literacy - Poor internet connectivity - Lack of smart devices and sensors - Limited access to modern farming technologies	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions: - Traditional farming methods - Weather forecast apps - Basic farm management software Pros: - Easy to use - Low initial cost - Basic information available Cons: - No AI-based recommendations - Limited data integration - Low prediction accuracy - No real-time optimization
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2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]

- Unpredictable weather conditions (High)
- Low crop yield due to poor planning (High)
- Excessive water and fertilizer usage (High)
- Pest and disease outbreaks (Medium)
- Unstable market prices (Medium)

9. PROBLEM ROOT / CAUSE [RC]

- Lack of real-time agricultural insights
- Inefficient resource management
- Climate variability
- Limited access to accurate farming data
- Poor decision-making due to insufficient analytics

7. BEHAVIOR
+ ITS INTENSITY [BE]

- Farmers rely on traditional practices (High)
- Decisions based on experience rather than data (High)
- Technology adoption is gradually increasing (Medium)
- Need quick and simple recommendations (High)

3. TRIGGERS TO ACT [TR]

- Unpredictable weather conditions
- Declining crop yield
- Rising farming costs
- Water scarcity
- Pest and disease outbreaks
- Need for higher profits and sustainable farming

10. YOUR SOLUTION [SL]

OptiCrop is an AI-powered smart agricultural production optimization engine that analyzes soil health, weather forecasts, crop conditions, and market trends to provide personalized recommendations for crop selection, irrigation scheduling, fertilizer usage, pest detection, and yield prediction. It helps farmers improve productivity, reduce costs, conserve resources, and make data-driven farming decisions.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Mobile application
- Web dashboard
- SMS/WhatsApp alerts
- Agricultural websites
- Social media (YouTube, Facebook)

4. EMOTIONS
BEFORE / AFTER [EM]

Before:

- Uncertainty about farming

OFFLINE

- Agricultural extension officers
- Farmer meetings and workshops

decisions

- Stress due to crop failures
- Financial insecurity
- Fear of unpredictable weather

After:

- Confidence in decision-making
- Increased productivity
- Better income and profitability
- Satisfaction with optimized resource usage

- Local agricultural cooperatives
 - Government agriculture offices
 - Training programs and field demonstrations
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